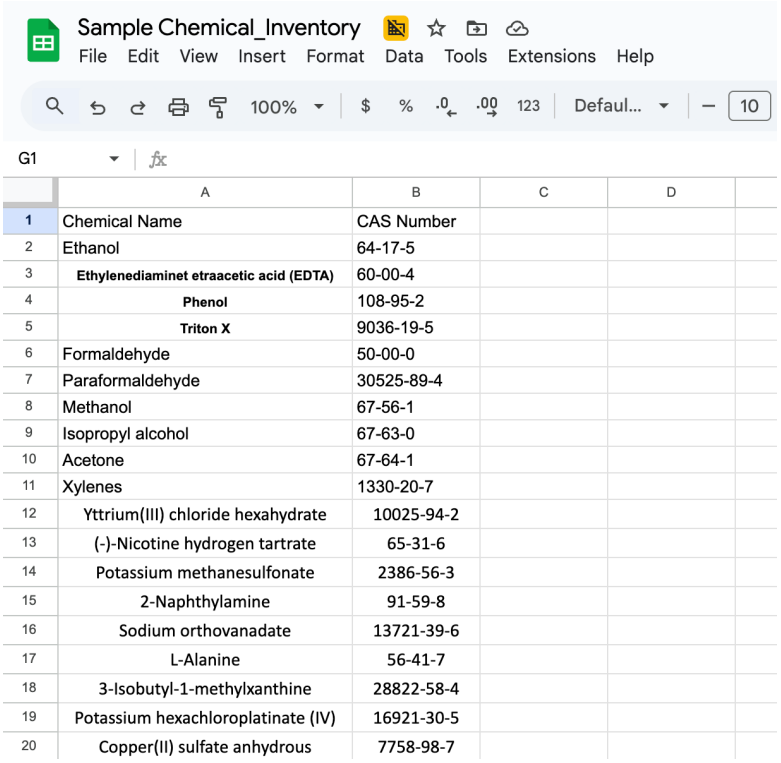
HazardPyMatch ReadMe

STEP 1 - Before you run HazardPyMatch

1. Choose your H-codes of interest, referring to the GHS Classification (Rev.10, 2023) Summary found here: https://pubchem.ncbi.nlm.nih.gov/ghs
2. Create a folder called ”your_source_folder” in MyDrive
3. Upload a chemical list, either a .xlsx or .csv file, to My Drive/your_source_folder
4. *Important notes:*
    1. The name of the document you upload must have “Chemical_Inventory” in the name e.g. Sample Chemical_Inventory.csv (**Screenshot 1**)
    2. Two of the column headers must be “Chemical Name” and ”CAS Number” (**Screenshot 1**)
    3. Limit one Chemical Name per row.
5. Create a folder called “your_protocols_folder” in MyDrive
6. Upload any protocol documents to this folder. The documents must be .pdf format

Proceed to STEP 2, next page…

Screenshot 1



	A	B	C	D
1	Chemical Name	CAS Number		
2	Ethanol	64-17-5		
3	Ethylenediaminet etraacetic acid (EDTA)	60-00-4		
4	Phenol	108-95-2		
5	Triton X	9036-19-5		
6	Formaldehyde	50-00-0		
7	Paraformaldehyde	30525-89-4		
8	Methanol	67-56-1		
9	Isopropyl alcohol	67-63-0		
10	Acetone	67-64-1		
11	Xylenes	1330-20-7		
12	Yttrium(III) chloride hexahydrate	10025-94-2		
13	(-)-Nicotine hydrogen tartrate	65-31-6		
14	Potassium methanesulfonate	2386-56-3		
15	2-Naphthylamine	91-59-8		
16	Sodium orthovanadate	13721-39-6		
17	L-Alanine	56-41-7		
18	3-Isobutyl-1-methylxanthine	28822-58-4		
19	Potassium hexachloroplatinate (IV)	16921-30-5		
20	Copper(II) sulfate anhydrous	7758-98-7		

STEP 2 – How to run HazardPyMatch

1. Upload HazardPyMatch.ipynb to My Drive/your_source_folder
2. Make sure the folder names in lines 8 and 14 in the first code block are correct (purple arrows **Screenshot 2**)
3. If you want to print all chemical inventory outputs at each intermediate step, set line 18 to ”True” (orange arrow **Screenshot 2**). Do not use quotation marks.

4. Runtime > Run all (**Screenshot 3**)
5. You will see a box appear (**Screenshot 4**) at the bottom of the first code block. Type in the H-codes corresponding to your hazard classification interest e.g. H200, H201, H360FD (**Screenshot 4**) and click Enter.
6. Refresh My Drive/your_source_folder for outputs!

Screenshot 2

Define I/O configuration and file system paths

```
[ ]
1  import os
2  import glob
3  from google.colab import drive
4  drive.mount('/content/drive')
5
6  # Define all person specific locations and file names
7  # Define source folder
8  source_folder = '/content/drive/My Drive/your_source_folder/'
9
10 # Define location of Chemical Inventory, which is an input list including 'Chemical Names' and 'CAS Numbers' column names
11 inventory_files = glob.glob(os.path.join(source_folder, '*Chemical_Inventory*.xlsx'))
12
13 # Define protocol folder
14 protocols_folder = os.path.join(source_folder, 'your_protocols_folder')
15
16 # = True if you want to print intermediate steps of chemical inventory dataframe for debugging.
17 # Set at = False as default
18 print_intermediate_steps = False
19
20 # Prompt the user to input relevant GHS codes
21 print("Enter relevant GHS codes separated by commas (e.g., H200, H201, H360FD):")
22 relevant_ghs_codes = input().strip().split(',')
```

Screenshot 3

HazardPyMatch.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Tex Run all ⌘/Ctrl+F9

Screenshot 4

```
Mounted at /content/drive
Enter relevant GHS codes separated by commas (e.g., H200,H201,H360FD):
H200,H201,H360FD
```

Example outputs from HazardPyMatch using the demo file Sample Chemical_Inventory.csv

A. List of Chemical Names lacking CAS Numbers. These can include proprietary reagents and also chemical solutions with names that are not explicitly recognized by PubChem, with an example shown below. The contentious user can search supplier SDS or identify the most likely CAS Number for the chemical name in this list, enter it in the input file and rerun the pipeline.

B. Output 1 is a Chemical Inventory containing only user chosen GHS H-Codes. Here displaying the first 10 chemical names. Note that due to threading implemented at a previous step, the order of this list may be different compared to below.

C. Output 2 includes plots and lists for final CAS Numbers in Protocols. The first file out is a bar chart showing the number of hazards identified in each protocol. Next, Pareto Analysis of the number of protocols containing specific Hazard CAS Numbers.

Proceed to Output 2 lists, next page…

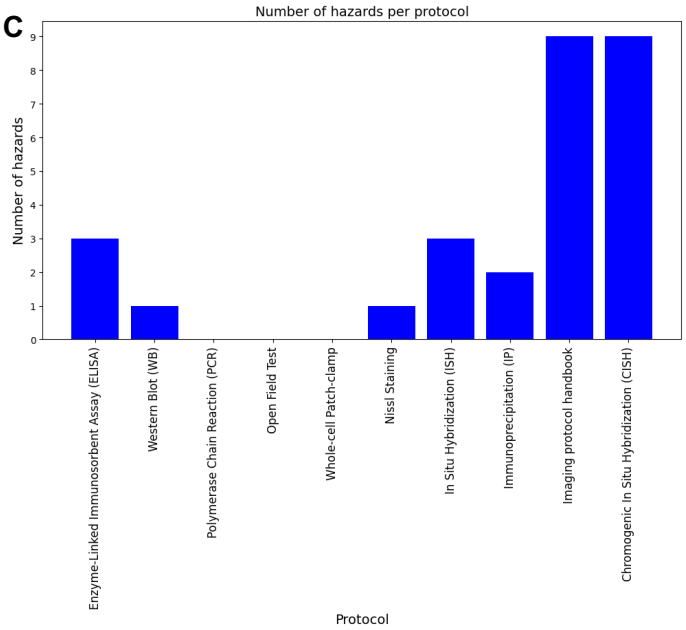
A

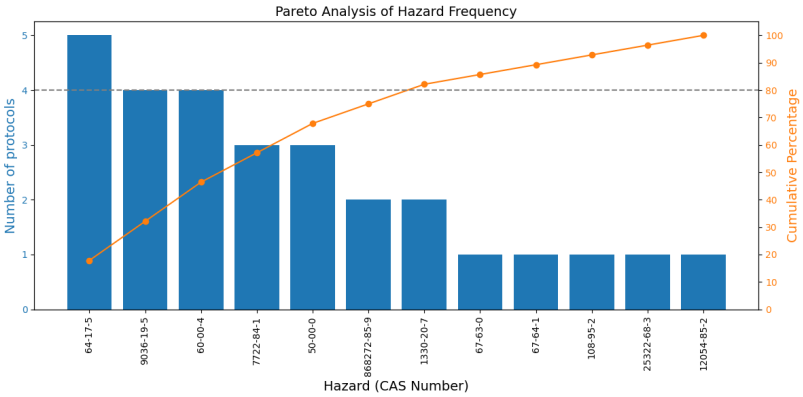
Chemical Name	CAS Number
Paraformaldehyde Aqueous Solution	

B

Chemical Name	CAS Number	PubChem ID	H Codes	H_codes_found	has_relevant	matched_relevant
Ethanol	64-17-5	702	H335: May c	['H225', 'H315', 'H319',	TRUE	['H340', 'H360']
Formaldehyde	50-00-0	712	H350: May c	['H220', 'H227', 'H280',	TRUE	['H341']
Methanol	67-56-1	887	H311: Toxic i	['H225', 'H301', 'H302',	TRUE	['H360']
Phenol	108-95-2	996	H311: Toxic i	['H301', 'H302', 'H311',	TRUE	['H340', 'H341', 'H360']
Phenol	108-95-2	7966	H316: Cause	['H302', 'H303', 'H312',	TRUE	['H361']
Paraformaldehyde	30525-89-4	712	H350: May c	['H220', 'H227', 'H280',	TRUE	['H341']
Ethylenediaminet etraacetic acid (EDTA)	60-00-4	6049	H335 (31.8%	['H302', 'H303', 'H312',	TRUE	['H361']
Isopropyl alcohol	67-63-0	3776	H335: May c	['H225', 'H303', 'H305',	TRUE	['H361']
Acetone	67-64-1	180	H335: May c	['H225', 'H305', 'H319',	TRUE	['H361']
Triton X	9036-19-5	24775	H314 (100%	['H302', 'H314', 'H361',	TRUE	['H361']

C





Example outputs from HazardPyMatch using the demo file Sample Chemical_Inventory.csv

C. **(continued)** The list output ”Hazards_in_Protocols” is the specific key for the Bar chart (above). Finally, we provide the specific details for the hazard chemical name or synonyms identified by the protocol search function in the document ”Protocol_Matched_Hazard_Details” (below). Here displaying the first 10 hazard synonyms found. Note that the pdf searching function gives false positives, for example, the word “Protection” identified in Enzyme-Linked Immunosorbent Assay (ELISA)_Thermo Scientific.pdf. ”Protection” was identified by the software because this word was included as a synonym in the PubChem ID entry 3776 for CAS Number 67-63-0, Isopropyl Alcohol. For complete thoroughness, the software pulls *all possible* synonyms for each PubChem ID and does not screen them. Users identify false positives using own discretion.

Protocol	Source	Hazards
Enzyme-Linked Immunosorbent Assay (ELISA)	Thermo Scientific	67-63-0, 64-17-5, 7722-84-1
Western Blot (WB)	BioRad	9036-19-5
Polymerase Chain Reaction (PCR)	The Jackson Laboratory	N/A
Open Field Test	Stanford University	N/A
Whole-cell Patch-clamp	Axol Bio	N/A
Nissl Staining	Cold Spring Harbor Protocols	64-17-5
In Situ Hybridization (ISH)	Abcam	60-00-4, 64-17-5, 50-00-0
Immunoprecipitation (IP)	Abcam	9036-19-5, 60-00-4
Imaging protocol handbook	ThermoFisher	868272-85-9, 67-64-1, 9036-19-5, 60-00-4, 64-17-5, 50-00-0, 7722-84-1, 108-95-2, 1330-20-7
Chromogenic In Situ Hybridization (CISH)	Springer Protocols	868272-85-9, 25322-68-3, 9036-19-5, 12054-85-2, 60-00-4, 64-17-5, 50-00-0, 1330-20-7, 7722-84-1

Protocol	Specific word pulled from pdf	Matched CAS Number	Matched PubChem ID	Matched GHS H Codes
Enzyme-Linked Immunosorbent Assay (ELISA)_Thermo Scientific.pdf	Protection	67-63-0	3776	H361
Enzyme-Linked Immunosorbent Assay (ELISA)_Thermo Scientific.pdf	Dynamic	64-17-5	702	H340;H360
Enzyme-Linked Immunosorbent Assay (ELISA)_Thermo Scientific.pdf	a peroxide	7722-84-1	784	H361
Western Blot (WB)_BioRad.pdf	Triton X	9036-19-5	24775	H361
Nissl Staining_Cold Spring Harbor Protocols.pdf	Ethanol	64-17-5	702	H340;H360
In Situ Hybridization (ISH)_Abcam.pdf	EDTA	60-00-4	6049	H361
In Situ Hybridization (ISH)_Abcam.pdf	ethanol	64-17-5	702	H340;H360
In Situ Hybridization (ISH)_Abcam.pdf	formaldehyde	50-00-0	712	H341
Immunoprecipitation (IP)_Abcam.pdf	Triton X	9036-19-5	24775	H361
Immunoprecipitation (IP)_Abcam.pdf	EDTA	60-00-4	6049	H361